

Executable Research Plan

Leakage-Safe Knowledge-Augmented Multimodal Learning for Tox21 Drug Toxicity Prediction

Stretch Target: Mean Test ROC-AUC ≥ 0.96

Scientific integrity statement: 0.96 একটি *stretch target*, guarantee নয়। Result তখনই valid ধরা হবে যখন frozen protocol, scaffold-based holdout, per-task masking, target-edge removal, no test tuning, এবং confidence interval সব শর্ত পূরণ হবে। কোনো leakage বা repeated test peeking-এর মাধ্যমে score বাড়ানো এই plan-এর অংশ নয়।

Prepared for: Drug Toxicity Prediction Thesis Project

Primary dataset: Tox21, 12 toxicity endpoints

Central model: Knowledge-Augmented Multimodal GPS Fusion Network

Validation: 75/25 scaffold holdout + 3-fold CV; top models confirmatory 5-fold

Evidence base: Thesis proposal, previously reviewed toxicity/ML papers, and Xie et al. (2025) ToxKG-GNN paper

Executive Summary

এই plan-এর উদ্দেশ্য শুধু একটি বড় AUC দেখানো নয়; বরং এমন একটি reproducible, leakage-safe এবং explainable system তৈরি করা যা Tox21-এর 12টি endpoint-এ নতুন molecular scaffolds-এর উপর robust prediction দিতে পারে। নতুন ToxKG-GNN paper দেখিয়েছে যে molecular structure-এর সাথে chemical-gene-pathway knowledge integrate করলে homogeneous GNN-এর তুলনায় বড় উন্নতি সম্ভব। Paper-এ GPS model-এর average AUC 0.911 এবং কিছু endpoint-এ 0.95-এর বেশি score report করা হয়েছে। আমাদের লক্ষ্য আরও কঠিন: **untouched scaffold test set-এ 12 endpoint-এর macro mean ROC-AUC ≥ 0.96** ।

≥ 0.96

Primary stretch target: mean test ROC-AUC

12

Jointly learned Tox21 endpoints

0

Allowed target-label leakage paths

Core technical hypothesis: Local molecular structure, global toxicological knowledge, pretrained SMILES semantics, and robust fingerprint features একসাথে task-conditioned gated fusion-এর মাধ্যমে combine করলে এবং প্রতিটি branch আলাদাভাবে pretrain ও validate করলে structure-only বা KG-only model-এর সীমাবদ্ধতা অতিক্রম করা সম্ভব।

Final proposed system

Branch A - Local Molecular Graph

AttentiveFP / MPNN, atom-bond features, edge-aware message passing, atom attention.

Branch B - Toxicological Knowledge Graph

GPS primary, HGT/R-GCN backup; chemical-gene-pathway relations only, target assay outcome edges removed.

Branch C - SMILES Language Model

ChemBERTa with staged unfreezing, randomized-SMILES augmentation, contextual embedding.

Branch D - Structural Feature Encoder

Five fingerprints + curated descriptors, fold-local variance filtering and SVD/PCA, DeepTox-style encoder.

Task-conditioned cross-modal attention -> gated residual fusion -> NR/SR shared trunks -> 12 task heads -> endpoint-wise weighted ensemble

Highest-impact upgrades for reaching the target

- Leakage-safe ToxKG:** target Tox21 active/inactive assay edges, direct label-derived features, and any post-split information are removed before graph construction.
- Cross-modal self-supervised pretraining:** graph-SMILES-KG representations aligned before supervised fine-tuning.
- Task-conditioned fusion:** each endpoint learns different weights for graph, KG, SMILES, and fingerprint evidence.
- Endpoint-specific out-of-fold ensemble:** ensemble weights are learned only from CV predictions, never from final test.

5. **Robust optimization:** masked class-balanced focal/asymmetric loss, balanced batching, EMA/SWA, mixed precision, and fold/seed ensembling.

Non-negotiable: যদি valid protocol-এ 0.96 না আসে, result manipulate করা হবে না। Best valid score, uncertainty, failure analysis এবং next-step evidence report করা হবে।

Table of Contents

1. Target Definition and Success Criteria	14. Training Schedule
2. Research Questions and Hypotheses	15. Hyperparameter Optimization
3. Frozen Evaluation Protocol	16. Hybrid Models and Ensemble
4. Data Audit and Chemical Curation	17. Evaluation and Statistics
5. Missing-Label Masking	18. Explainability and Fidelity
6. Toxicological Knowledge Graph	19. Calibration, AD and Conformal Prediction
7. Leakage Audit	20. Ablation and Robustness
8. Data Cohorts and Splits	21. Experiment Matrix
9. Molecular Representations	22. Stage Gates and Timeline
10. Self-Supervised Pretraining	23. Folder Architecture and Reproducibility
11. Model Suite	24. Risk Register
12. Proposed Multimodal Architecture	25. Final Acceptance Checklist
13. Imbalance Handling	

1. Target Definition and Success Criteria

1.1 Primary target

Primary success criterion: The final frozen ensemble must achieve **macro mean ROC-AUC ≥ 0.960** across the 12 Tox21 endpoints on a single untouched scaffold-based test set, with all missing labels masked endpoint-wise.

1.2 Why the target is difficult

Recent knowledge-graph GNN evidence reports average AUC around 0.911, while individual endpoints can reach approximately 0.956-0.966. Therefore, a **mean** AUC of 0.96 is substantially harder than achieving 0.96 on one endpoint. The plan must improve weaker endpoints, not only maximize already easy tasks.

1.3 Valid score definition

Requirement	Mandatory rule	Invalid shortcut
Split	Bemis-Murcko scaffold holdout; final test frozen before modeling	Random split used as primary result
Feature fitting	Scaler, PCA/SVD, threshold, ensemble weights fit inside training/CV only	Fit on full dataset
Knowledge graph	Target assay outcome edges and label-derived features removed	Active/inactive target labels present in graph
Model selection	Based on CV/OOF performance only	Repeatedly choose model using test score
Reporting	Mean, per-endpoint scores, 95% CI, seed/fold variability	Report only best endpoint or best fold

1.4 Secondary acceptance criteria

- CV mean AUC close to test AUC; large CV-test gap triggers audit.
- At least 10/12 endpoints should improve over the strongest same-split baseline.
- No endpoint should gain AUC only by collapsing minority-class detection; PR-AUC, BAC and F1 are diagnostic gates.
- Bootstrap 95% CI should support the claimed improvement.
- Model must generate atom-level and gene/pathway-level explanations with fidelity tests.

2. Research Questions and Hypotheses

ID	Research question	Testable hypothesis
RQ1	Does leakage-safe external toxicological knowledge improve scaffold generalization?	GPS/HGT with chemical-gene-pathway relations improves mean AUC over AttentiveFP alone on the identical KG-matched cohort.
RQ2	Does cross-modal pretraining improve performance beyond simple feature concatenation?	Graph-SMILES-KG contrastive alignment yields higher OOF AUC and lower fold variance.
RQ3	Do endpoint-conditioned gates outperform a single shared fusion weight?	Task-conditioned gates improve difficult endpoints such as rare-positive tasks without harming easy endpoints.
RQ4	Can ensemble diversity lift mean AUC toward 0.96?	OOF-optimized endpoint-wise blending of GPS fusion, AttentiveFP, ChemBERTa and DeepTox improves mean test AUC over every individual model.
RQ5	Are explanations mechanistically meaningful?	Removing top-ranked atoms/relations produces larger prediction drops than removing random atoms/relations.

2.1 Optimization objective

Hyperparameter search will not optimize only mean AUC. It will use a robustness-aware objective to prevent one or two easy endpoints from dominating.

```
J = mean_endpoint_AUC
  - 0.15 * std(endpoint_AUC)
  - 0.10 * max(0, 0.82 - min_endpoint_AUC)
  - 0.05 * calibration_penalty
```

The exact coefficients may be tuned on CV only. Final test remains untouched.

3. Frozen Evaluation Protocol



3.1 Split policy

- **Primary:** 75% training pool and 25% final test using Bemis-Murcko scaffold groups.
- **Mandatory model development:** 3-fold scaffold-aware CV inside the 75% training pool.
- **Confirmatory:** top two architectures rerun with 5-fold CV if compute permits.
- **Test access:** one formal evaluation after architecture, hyperparameters, thresholds and ensemble weights are frozen.

3.2 Split quality checks

- Zero scaffold overlap between train and test.
- Endpoint label coverage checked in every fold.
- Rare-positive endpoints must contain enough positives in validation/test; if impossible, use group-aware iterative balancing while preserving scaffold separation.

- Store molecule IDs, canonical SMILES, scaffold IDs and split assignment in a versioned CSV.

3.3 Comparison cohorts

Cohort	Purpose	Models
Full structure cohort	Use all chemically valid Tox21 molecules	DeepTox, CapsNet, AttentiveFP/MPNN, ChemBERTa, DenseNet, optional 3D EGNN
KG-matched cohort	Fair comparison of KG and non-KG models	GCN, GAT, R-GCN, HGT, GPS, AttentiveFP, ChemBERTa, multimodal fusion
External validation cohort	Generalization beyond Tox21	Final frozen model only; overlapping molecules removed

4. Data Audit and Chemical Curation

4.1 Initial audit outputs

Audit block	Required output
Dataset shape	Molecules, columns, endpoint names, label dtypes
Missingness	Available/missing labels per endpoint and per molecule
Class imbalance	Toxic, non-toxic, positive rate, negative:positive ratio
Chemical validity	Invalid SMILES, salts, mixtures, sanitize failures
Identity	Canonical duplicates, InChIKey duplicates, conflicting labels
Chemical space	Scaffold frequency, MW, LogP, TPSA, rings, SMILES length

4.2 Chemical standardization pipeline



4.3 Duplicate and conflict policy

- Identical canonical molecule and consistent labels: merge to one record.
- Identical molecule with conflicting endpoint labels: mark only that endpoint as uncertain/missing; do not delete all other valid labels.
- Keep an audit table with original row IDs and all transformations.
- All molecular representations must be generated from the same standardized molecule object.

Expected benefit: consistent chemical identity reduces representation noise across graph, SMILES, image, fingerprint and KG branches.

5. Per-Task Missing-Label Masking

Tox21 is partially labeled. Missing does not mean non-toxic. The model will preserve all partially labeled molecules and compute loss only for observed endpoint labels.

```
Y[i,t] = observed binary label, or dummy 0 when missing
M[i,t] = 1 if label is observed, else 0
L_raw  = focal_or_asymmetric_BCE(Y, P, class_weight_t)
L_masked = sum(L_raw * M) / max(sum(M), 1)
```

5.1 Implementation rules

- Mask is applied in training loss, validation metrics, CV metrics and test metrics.
- Class weights are computed from the training portion of each fold only.
- Endpoint metrics are calculated only if both classes are present.

- No missing label is imputed to 0 or 1.
- For hybrid per-endpoint classifiers, train only on rows with an observed label for that endpoint.

6. Toxicological Knowledge Graph

6.1 Sources and entities

Primary sources: ComptoxAI, PubChem, ChEMBL and Reactome. Optional enrichment: AOP-Wiki, CTD, external non-target ToxCast assays, literature-derived relations after provenance filtering.

Node types

Chemical, Gene, Protein, Pathway, Assay, Key Event, Molecular Initiating Event, Adverse Outcome.

Core relations

Chemical-binds-Gene, Chemical-increases/decreases-expression, Gene-in-Pathway, Gene-interacts-Gene, pathway hierarchy.

6.2 CID and identity mapping



All API results are cached. Mapping confidence, ambiguous matches and failed mappings are stored in [datasets/cache/pubchem_mapping.csv](#) .

6.3 KG quality controls

- Every edge stores source database, source version/date, evidence type and confidence.
- Remove duplicates, self-loops without meaning, orphan nodes and obsolete identifiers.
- Separate relation types; do not collapse all edges into one homogeneous relation.
- Keep a frozen KG snapshot for reproducibility.

7. Mandatory Leakage Audit

Critical risk: Chemical-HAS-ACTIVE-ASSAY or Chemical-HAS-INACTIVE-ASSAY edges may directly encode the target Tox21 outcome. If retained, AUC can become artificially high and scientifically invalid.

7.1 Edges/features to remove

- All edges containing the same Tox21 endpoint outcomes being predicted.
- Any node feature generated from the 12 target labels.
- Any assay summary feature calculated using validation or test labels.
- Any post-split statistics that use the full dataset.

7.2 Automated leakage tests

Test	Procedure	Pass condition
Relation-name audit	Search target assay names and active/inactive labels in KG schema	No target outcome edge remains
Feature provenance audit	Trace every feature to source and timestamp	No target-derived feature
Label permutation test	Shuffle train labels and retrain small model	AUC approximately chance
Target-edge ablation	Compare safe KG vs intentionally removed target edges	Safe model remains plausible; suspicious jumps trigger rejection
Split overlap audit	Check molecule, InChIKey and scaffold overlap	Zero prohibited overlap

8. Molecular Representations

8.1 Local molecular graph

Atom features: atomic number, degree, formal charge, hybridization, aromaticity, chirality, hydrogen count, valence, ring membership, atomic mass and selected donor/acceptor flags.

Bond features: bond type, direction, conjugation, ring status, stereo configuration and bond order.

8.2 Fingerprints and descriptors

Family	Features	Processing
Fingerprints	Morgan/ECFP4, MACCS, Atom-Pair, RDKit/FP2, optional topological torsion	VarianceThreshold -> TruncatedSVD; fit inside fold only
Descriptors	MW, LogP, TPSA, HBD, HBA, rotatable bonds, ring counts, charge, heavy atoms, fraction Csp3, etc.	Median imputation -> robust/standard scaling -> PCA if useful; fit inside fold only

8.3 SMILES representation

- Canonical SMILES for deterministic validation/test.
- Randomized SMILES augmentation during training only.

- Test-time augmentation may average canonical plus a small fixed set of randomized SMILES generated without labels.

8.4 Optional 2D and 3D representations

- **DenseNet121:** standardized 224x224 RDKit depictions; secondary branch/ablation only unless it adds measurable OOF value.
- **3D EGNN:** ETKDG conformer ensemble, geometry optimization and equivariant encoder; added only after core model is stable and if OOF gain exceeds the predefined gate.

9. Self-Supervised and Cross-Modal Pretraining

Why this is necessary: Tox21 is small and imbalanced relative to the capacity of GPS, HGT and ChemBERTa. Pretraining is the strongest scientifically legitimate lever for approaching the 0.96 target without using test labels.

9.1 Branch-specific pretraining

Branch	Pretraining tasks	Data
Molecular graph	Masked atom/bond prediction, graph contrastive learning, molecular property auxiliary tasks	External unlabeled molecules from ChEMBL/ PubChem or available pretrained checkpoints
Knowledge graph	Relation prediction, link prediction, masked node/ edge type prediction	Leakage-safe ToxKG snapshot
SMILES	Use pretrained ChemBERTa; optional domain-adaptive masked language modeling	Unlabeled standardized SMILES only
Fingerprint encoder	Denoising/autoencoding, descriptor reconstruction	Training molecules and external unlabeled molecules

9.2 Cross-modal alignment

Positive pair: same molecule across graph, SMILES, fingerprint and KG views

Negative pair: different molecules, scaffold-aware hard negatives

```
Loss = supervised_masked_loss  
      + lambda1 * graph_SMILES_contrastive  
      + lambda2 * graph_KG_contrastive  
      + lambda3 * cross_modal_consistency
```

9.3 Pretraining safeguards

- No target Tox21 test labels or target assay outcome edges.
- External molecules overlapping final test may be used only in fully unsupervised pretraining if no target labels are used; for the strictest experiment, exclude overlaps and report both settings.
- Pretraining checkpoints and data versions are frozen before supervised fine-tuning.

10. Model Suite and Priority

Priority	Model	Role	Keep for final ensemble?
1	GPS toxicological KG encoder	Global mechanism-aware representation	Yes
2	AttentiveFP / MPNN	Local atom-bond structure and atom-level explanation	Yes
3	ChemBERTa multitask	SMILES semantics and complementary errors	Yes if diverse
4	DeepTox-style multitask DNN	Strong fingerprint/descriptor baseline	Likely

5	HGT and R-GCN	Heterogeneous KG alternatives and endpoint specialists	Endpoint-dependent
6	Multitask CapsNet	Rare/minority pattern specialist	Only if OOF gain
7	DenseNet121	Image-based diversity and Grad-CAM	Only if incremental value
8	3D EGNN	Advanced geometry branch	Optional
Baseline	RF, SVM-RBF, XGBoost	Thesis comparison and hybrid heads	Not central

11. Proposed Architecture v2

11.1 Leakage-Safe Knowledge-Augmented Multimodal GPS Network

A. AttentiveFP/MPNN local encoder

4-5 message-passing layers, hidden 256, edge-aware attention, graph pooling, atom importance.

B. GPS/HGT KG encoder

4 GPS blocks: GINE local message passing + global Transformer, relation embeddings, chemical-gene-pathway context.

C. ChemBERTa encoder

Pretrained transformer, staged unfreezing, pooled embedding, randomized-SMILES consistency.

D. Fingerprint/descriptor encoder

Fold-local SVD/PCA, DeepTox-style residual MLP, 256-dimensional structural embedding.

Projection to 256d -> 2-layer cross-modal Transformer -> modality dropout -> endpoint-conditioned gates -> NR shared trunk / SR shared trunk -> 12 calibrated sigmoid heads

11.2 Task-conditioned gating

একই modality সব endpoint-এ সমান useful নয়। প্রতিটি endpoint-এর জন্য learned task embedding ব্যবহার করে gate weight নির্ধারণ করা হবে।

```
for endpoint t:
    q_t = learned_task_embedding[t]
    alpha_t = softmax(Gate(q_t, z_graph, z_kg, z_smiles, z_fp))
    z_t = sum_m alpha_t[m] * Project_m(z_m)
    p_t = sigmoid(Head_t(z_t))
```

11.3 NR/SR hierarchical sharing

- Shared base across all endpoints.
- Nuclear receptor (NR) shared adapter.
- Stress response (SR) shared adapter.
- Final endpoint-specific heads.

This structure balances transfer learning with task-specific specialization.

11.4 Modality dropout

Training-এর সময় random branch dropout করা হবে, যাতে model শুধু KG বা শুধু ChemBERTa-এর উপর over-rely না করে এবং missing-KG molecules handle করতে পারে।

12. Imbalance Handling

12.1 Primary loss

Two candidates will be compared on CV:

1. **Masked class-balanced focal BCE** using effective number of samples.

2. **Masked asymmetric loss** for multi-label classification, reducing easy-negative dominance.

```
w_pos[t] = (1 - beta) / (1 - beta ** n_pos[t])  
w_neg[t] = (1 - beta) / (1 - beta ** n_neg[t])  
loss = masked_asymmetric_or_focal_BCE(y, p, w_pos, w_neg)
```

12.2 Additional controls

- Positive-aware batch sampler, while preserving unbiased validation/test.
- Endpoint-wise loss normalization so dense-label tasks do not dominate.
- Curriculum: start with weighted BCE, then switch to focal/asymmetric loss after stable learning.
- SMOTE is prohibited on raw graph, raw SMILES and images; optional only on frozen embeddings for hybrid classifiers.

13. Training Schedule

13.1 Three-stage training

Stage A - Unimodal competence

Train/fine-tune each branch separately using the same folds. Save OOF predictions and embeddings.

Stage B - Frozen-encoder fusion

Freeze branch encoders; train cross-modal attention, gates and task heads. This stabilizes fusion.

Stage C - Partial joint fine-tuning

Unfreeze top layers with discriminative learning rates and strong early stopping.

13.2 Initial hyperparameters

Component	Initial setting	Search range
Graph/KG hidden dimension	256	192, 256, 384
GPS/HGT layers	4	3-6
Attention heads	8	4, 8
Dropout	0.25	0.10-0.45
Optimizer	AdamW	fixed
LR - new heads	3e-4	1e-4 to 1e-3
LR - pretrained encoders	1e-5 to 3e-5	5e-6 to 5e-5
Weight decay	1e-5	1e-6 to 1e-3
Batch size	32/64	hardware-dependent
Max epochs	120	early stopping required
Patience	12	8-20

13.3 Stability and speed

- Automatic mixed precision (AMP).
- Gradient clipping at 1.0.
- Exponential moving average (EMA) or stochastic weight averaging (SWA).
- Cosine decay with warmup or ReduceLROnPlateau.
- Gradient accumulation if GPU memory is limited.
- Deterministic seeds for audit; additionally run 3 seeds for final candidates.

14. Hyperparameter Optimization

Use Optuna with pruning. The search is staged to avoid wasting compute.

Stage	Trials	What is tuned	Selection
Coarse branch search	20-30/ model	depth, hidden size, dropout, LR, loss type	3-fold OOF objective

Fusion search	30-50	cross-modal layers, gate temperature, modality dropout, auxiliary loss weights	robust objective J
Local refinement	10-20	narrow ranges around best config	mean AUC + variance

Test-set rule: Optuna objective, pruning and trial selection never use final test predictions.

15. Hybrid Models and Embedding Heads

Deep encoders can produce strong fixed embeddings for small-data classifiers. These models are secondary, but may improve endpoint-specific performance.

Hybrid	Purpose	Condition to retain
GPS + AttentiveFP embedding -> SVM-RBF	Nonlinear classifier on combined mechanism/structure embedding	OOF AUC gain ≥ 0.003 and diverse errors
GPS embedding -> XGBoost	Endpoint specialist	Improves at least two weak endpoints
ChemBERTa embedding -> SVM-RBF	Stable small-data sequence classifier	Adds ensemble diversity
CapsNet embedding -> SVM/RF	Minority-class specialist	Improves PR-AUC/BAC without AUC loss
DenseNet embedding -> SVM/RF	Image-based diversity	OOF improvement justifies compute

16. Ensemble Strategy

16.1 Two-level ensemble

1. **Within-model ensemble:** 3 folds x 3 seeds, averaged probabilities.
2. **Across-model ensemble:** endpoint-wise OOF-optimized weights across diverse models.

16.2 Candidate members

Multimodal GPS fusion AttentiveFP HGT/R-GCN specialist ChemBERTa DeepTox DNN Best hybrid

16.3 Weight learning

```
For each endpoint t:
  Input: out-of-fold probabilities from candidate models
  Optimize non-negative weights  $w_t$ ,  $\sum(w_t)=1$ 
  Objective: maximize OOF ROC-AUC
    - variance penalty
    - calibration penalty
  Freeze  $w_t$  before final test prediction
```

16.4 Diversity gate

A model is included only if it improves OOF ensemble AUC or reduces correlated errors. High individual AUC alone is not enough.

16.5 Test-time augmentation

- ChemBERTa: average canonical and fixed randomized-SMILES views.
- Graph: average only deterministic or carefully validated augmentation views.
- 3D: average multiple conformers if 3D branch is retained.

17. Evaluation and Statistical Analysis

17.1 Headline metrics

Metric	Role	Report
Macro mean ROC-AUC	Primary selection and target metric	Main table
Accuracy	User-requested secondary metric	Main table
PR-AUC	Minority toxic-class quality	Diagnostic/supplementary
Balanced Accuracy	Class-imbalance robustness	Diagnostic/supplementary
F1, MCC, Sensitivity, Specificity	Failure-mode diagnosis	Supplementary

17.2 Threshold policy

- ROC-AUC uses probabilities and no threshold.
- Accuracy/F1/BAC thresholds are selected per endpoint from OOF predictions only.
- Threshold criterion is predeclared (e.g., Youden J, balanced accuracy or cost-sensitive utility).
- Thresholds are frozen before test.

17.3 Uncertainty and significance

- Bootstrap 95% confidence intervals, stratified by endpoint where feasible.
- Paired bootstrap or DeLong-style comparison between proposed model and strongest same-split baseline.
- Report fold and seed mean +/- standard deviation.
- Do not directly claim superiority over literature results obtained on different subsets/splits; provide contextual comparison only.

17.4 Required final tables

1. Per-model: CV mean AUC, CV SD, test mean AUC, test accuracy.
2. Per-endpoint: AUC, accuracy, positive count, negative count, missing count.
3. Ensemble composition and endpoint-specific weights.
4. Ablation results and incremental gain.
5. Calibration, applicability-domain and conformal coverage.

18. Explainability and Fidelity

18.1 Three explanation levels

Level	Method	Output
Atom/bond	AttentiveFP/GAT attention, Integrated Gradients, GNNExplainer-style subgraph analysis	Highlighted toxicophore atoms and bonds

Biological mechanism	GPS/HGT relation attention and path extraction	Chemical -> gene -> pathway -> endpoint explanation
Feature	SHAP on fingerprint/descriptor encoder or hybrid head	Descriptor/substructure contribution

18.2 Fidelity tests

- Top-k atom deletion vs random atom deletion.
- Top-k KG edge/path removal vs random relation removal.
- Comprehensiveness: prediction drop after removing important evidence.
- Sufficiency: prediction retained using only selected evidence.
- Stability across folds/seeds.

Acceptance: An attention heatmap alone is not treated as proof. Explanations must pass perturbation-based fidelity tests.

19. Calibration, Applicability Domain and Conformal Prediction

19.1 Calibration

- Brier score, Expected Calibration Error and reliability diagrams.
- Temperature scaling or isotonic calibration fitted on OOF/calibration data only.
- Report calibrated and uncalibrated probabilities.

19.2 Applicability domain (AD)

Each prediction receives a reliability flag using multiple distances:

- Maximum Tanimoto similarity to training molecules.
- Scaffold novelty.
- Distance in learned multimodal embedding space.
- KG neighborhood coverage and relation completeness.

19.3 Conformal prediction

Use a calibration split/OOF conformal method to output prediction sets:

```
{Toxic}          -> high-confidence toxic
{Non-toxic}      -> high-confidence non-toxic
{Toxic, Non-toxic} -> uncertain; defer to experiment
{} (if method permits) -> out-of-distribution / review
```

19.4 Why these do not directly raise AUC

Calibration, AD and conformal prediction may not increase ROC-AUC, but they are essential for trustworthy deployment and prevent overclaiming.

20. Ablation and Robustness Studies

Ablation	Question answered
Fingerprint DNN only	How much does handcrafted structure explain?
AttentiveFP only	Value of local molecular graph
GPS KG only	Value of biological mechanism knowledge
ChemBERTa only	Value of SMILES language representation
Graph + KG	Core knowledge augmentation benefit
Graph + ChemBERTa	Structure-language complementarity
KG + ChemBERTa	Mechanism-language complementarity
All modalities	Full fusion benefit
Without relation types	Are heterogeneous semantics necessary?
Without pretraining	Contribution of self-supervised learning

Without class balancing	Impact of imbalance treatment
Without task-conditioned gates	Value of endpoint-specific fusion
Full model + ensemble	Final gain from model diversity

20.1 Robustness tests

- Three random seeds for final candidates.
- Alternative scaffold split seed/group assignment.
- KG edge dropout and missing-KG simulation.
- Label noise sensitivity.
- Rare-endpoint stress test.
- External dataset validation after overlap removal.

21. Experiment Matrix

Experiments are executed in a funnel. Expensive models proceed only after passing prior gates.

Phase	Experiment	CV	Go/No-Go gate
P0	Data curation + leakage audit	N/A	Zero target leakage; split frozen
P1	DeepTox DNN, RF, SVM-RBF, XGBoost	3-fold	Reproducible baseline
P2	AttentiveFP/MPNN, ChemBERTa	3-fold	At least one strong structural model
P3	R-GCN, HGT, GPS on safe KG	3-fold	KG improves same-cohort baseline
P4	Graph-KG fusion	3-fold	OOF gain ≥ 0.005 over best branch
P5	Add ChemBERTa + fingerprint encoder	3-fold	Each added modality provides incremental value or diversity
P6	Cross-modal pretraining + joint fine-tuning	3-fold	Lower variance and higher robust objective
P7	Endpoint-wise ensemble	OOF	Target approach; no test access
P8	Top two confirmatory 5-fold + 3 seeds	5-fold	Stable ranking
P9	One-time final scaffold test	Frozen	Primary target assessment
P10	External validation + XAI + uncertainty	Frozen	Thesis-ready evidence

21.1 Performance ladder

The exact values cannot be promised. The practical strategy is to earn improvement in layers: data quality -> strong branch models -> safe KG -> pretraining -> task-conditioned fusion -> endpoint-wise ensemble. Every gain must be validated OOF before it is accepted.

22. Stage Gates and Timeline

Week 1 - Protocol freeze

Data audit, chemical cleaning, split generation, leakage checklist, repository setup.

Week 2 - KG snapshot

CID mapping, relation filtering, target-edge removal, PyG HeteroData export.

Weeks 3-4 - Branch baselines

DeepTox, AttentiveFP/MPNN, ChemBERTa, R-GCN/HGT/GPS.

Weeks 5-6 - Pretraining and fusion

Graph/KG pretraining, cross-modal alignment, frozen fusion, partial joint fine-tuning.

Week 7 - HPO

Optuna coarse and local refinement; prune weak branches.

Week 8 - Ensemble

OOF predictions, endpoint-wise weights, seed/fold ensemble.

Week 9 - Confirmatory runs

Top two 5-fold/3-seed verification, calibration and uncertainty.

Week 10 - Final test and reporting

One-time test, bootstrap CI, XAI, ablations, figures, model card and thesis tables.

22.1 Gate thresholds

Gate	Pass criterion	Action if failed
Data gate	No leakage, valid masks, reproducible split	Stop modeling and fix data
KG gate	GPS/HGT beats same-cohort non-KG baseline	Audit mapping, relations and leakage-safe context
Fusion gate	OOF gain ≥ 0.005 or meaningful diversity	Remove weak modality
Stability gate	Low fold/seed variance; no endpoint collapse	Regularize, rebalance, simplify model
Target gate	Frozen test mean AUC ≥ 0.96	Report honest result; do not retune on test

23. Folder Architecture and Reproducibility

```
D:\Drug_Toxicity\  
|  
|-- datasets\  
|   |-- raw\tox21.csv  
|   |-- processed\tox21_curated.parquet  
|   |-- splits\scaffold_split_v1.csv  
|   |-- cache\pubchem_mapping.csv  
|   |-- external\  
|  
|-- knowledge_graph\  
|   |-- raw\  
|   |-- processed\toxkg_safe_v1\  
|   |-- provenance\edge_sources.parquet  
|  
|-- notebooks\  
|   |-- 00_environment_and_paths.ipynb  
|   |-- 01_data_audit_and_curation.ipynb  
|   |-- 02_kg_construction_and_leakage_audit.ipynb  
|   |-- 03_branch_models.ipynb  
|   |-- 04_pretraining_and_fusion.ipynb  
|   |-- 05_hpo_and_ensemble.ipynb  
|   |-- 06_final_evaluation_and_xai.ipynb  
|  
|-- src\  
|   |-- data\  models\  losses\  training\  evaluation\  xai\  
|  
|-- models\  
|-- outputs\  
|-- figures\  
|-- logs\  
|-- configs\  
|-- requirements\  
|-- reports\
```

23.1 Reproducibility artifacts

- ✓ Random seeds
- ✓ Environment lock file
- ✓ Data and KG version hashes
- ✓ Split indices and scaffold IDs
- ✓ Preprocessing objects
- ✓ Config YAML for every run
- ✓ Model checkpoints
- ✓ OOF and test predictions
- ✓ Metrics JSON/CSV
- ✓ Training logs
- ✓ Explainability artifacts
- ✓ Model card and limitations

23.2 Colab compatibility

- Use a path resolver that detects Windows local path vs Google Drive mount.
- Separate installation cell for RDKit, PyTorch Geometric, Transformers, Optuna and explainability packages.
- Cache expensive PubChem/KG preprocessing outputs in Drive.
- Provide a low-memory mode with frozen encoders and smaller batch sizes.

24. Compute and Software Stack

Layer	Recommended stack
Core	Python, PyTorch, PyTorch Geometric, RDKit, scikit-learn, pandas, NumPy
Transformers	Hugging Face Transformers / ChemBERTa checkpoint
Optimization	Optuna, AMP, TensorBoard or Weights & Biases
Explainability	Captum, SHAP, graph explanation utilities
KG storage	Neo4j for inspection; export frozen edge/node tables to PyG HeteroData for training
Recommended compute	24-48 GB GPU for full multimodal model; 64 GB system RAM; adequate SSD cache
Fallback	Colab Pro/Pro+ or freeze heavy encoders and train fusion heads

25. Risk Register

Risk	Impact	Mitigation
Target leakage through KG assay edges	Artificial 0.95-0.99 AUC	Relation removal, provenance audit, permutation test
Scaffold split causes rare endpoint scarcity	Unstable AUC	Group-aware iterative assignment, report support counts, confirmatory resampling
Multimodal overfitting	High CV-test gap	Pretrain, freeze-first training, modality dropout, weight decay, simplify fusion
KG coverage missing for many molecules	Reduced cohort and bias	Two-cohort analysis, missing-modality mask, modality dropout, coverage report
Compute exhaustion	Incomplete experiments	Stage gates, Optuna pruning, cache embeddings, stop weak branches early
High accuracy but poor toxic recall	Unsafe conclusions	PR-AUC/BAC/F1 diagnostics and class-balanced loss
Literature comparison is not protocol-matched	Overclaiming	Use same-split internal baselines; label external comparisons as contextual
0.96 target not reached	Project expectation risk	Report honest best valid score, CIs, bottleneck analysis and evidence-based next steps

26. What Must Never Be Done

- ✗ Use the same 12 Tox21 target labels as biological input features.
- ✗ Keep target active/inactive assay edges in the knowledge graph.
- ✗ Fit scaler, PCA/SVD, feature selection or class weights on the full dataset.
- ✗ Tune hyperparameters, thresholds or ensemble weights on final test predictions.
- ✗ Report the best fold, best seed or best endpoint as overall performance.
- ✗ Repeatedly peek at the test set until AUC reaches 0.96.
- ✗ Claim superiority over a paper without matching dataset, split and protocol.
- ✗ Hide failed endpoints, uncertainty, or class-support counts.

27. Final Acceptance Checklist

- ☐ Chemical curation audit complete
- ☐ Duplicate conflict policy applied
- ☐ Per-task mask validated
- ☐ Scaffold test frozen
- ☐ Zero molecule/scaffold leakage
- ☐ Target KG edges removed
- ☐ Feature provenance recorded
- ☐ Branch baselines reproduced
- ☐ GPS/HGT same-cohort comparison complete
- ☐ Pretraining ablation complete

- ☐ Fusion ablation complete
- ☐ OOF ensemble weights frozen
- ☐ Top models rerun with 3 seeds
- ☐ Final test evaluated once
- ☐ 95% CI reported

- ☐ AUC + Accuracy main tables ready
- ☐ PR-AUC/BAC/F1 diagnostics ready
- ☐ XAI fidelity tests complete
- ☐ Calibration/AD/conformal outputs ready
- ☐ Model card and limitations ready

28. Final Decision and Execution Order

Central model: GPS knowledge graph + AttentiveFP molecular graph + ChemBERTa + multi-fingerprint encoder -> leakage-safe cross-modal attention -> endpoint-conditioned gated fusion -> hierarchical NR/SR multitask heads -> OOF-optimized endpoint-wise ensemble.

Exact execution order

1. Freeze scientific protocol, metric definition and test-access policy.
2. Curate chemicals and build endpoint masks.
3. Create and freeze scaffold split.
4. Map PubChem IDs and build provenance-aware safe ToxKG.
5. Run leakage audits and label permutation test.
6. Train DeepTox, AttentiveFP/MPNN and ChemBERTa branch baselines.
7. Train R-GCN, HGT and GPS on the identical KG-matched cohort.
8. Pretrain graph/KG branches and align graph-SMILES-KG embeddings.
9. Train graph-KG fusion; retain only if OOF gain passes gate.
10. Add ChemBERTa and fingerprint branches incrementally, not all at once.
11. Run task-conditioned fusion HPO using robust OOF objective.
12. Generate fold/seed OOF predictions and learn endpoint-wise ensemble weights.
13. Run confirmatory top-model experiments and freeze all settings.
14. Evaluate the final test once; calculate CIs and diagnostics.
15. Complete XAI fidelity, calibration, applicability-domain and conformal analysis.
16. Publish reproducibility package, model card, limitations and thesis-ready tables.

Interpretation of the 0.96 target

Target, not promise: The plan is specifically optimized to maximize the probability of achieving a valid mean ROC-AUC ≥ 0.96 . However, the final score is determined by data quality, split difficulty, KG coverage, endpoint imbalance and true generalization. Scientific success means the highest reproducible, leakage-free result under the frozen protocol - not a manufactured number.

Final deliverables

- Curated Tox21 dataset and split manifest.
- Leakage-safe ToxKG snapshot with provenance.
- Reusable PyTorch/PyG training pipeline.
- Branch models, proposed multimodal model and final ensemble.
- OOF/test predictions and complete metrics.
- Thesis-quality figures, ablations, confidence intervals and explanation maps.
- Applicability-domain and conformal-confidence outputs.
- Model card, reproducibility checklist and documented limitations.

References and Evidence Base

1. Xie, J.; Liu, W.; Hu, W.; Ouyang, M.; Huang, T. *Graph Neural Network-Based Toxicity Prediction by Integrating Molecular Fingerprints and Knowledge Graph Features*. *Toxics*, 2025, 13, 953.
2. Thesis Proposal: *Drug Toxicity Prediction*, CSE478, Gopalganj Science and Technology University, 2025.
3. Mayr, A. et al. *DeepTox: Toxicity Prediction Using Deep Learning*.
4. Multitask Capsule Network study for imbalanced Tox21 toxicity prediction.
5. Cremer, J. et al. *Equivariant Graph Neural Networks for Toxicity Prediction*. 2023.
6. Popescu, E.; Groza, A.; Cernat, A. *Combining Deep Learning and Explainable AI for Toxicity Prediction of Chemical Compounds*.
7. Sharma, B. et al. *Accurate Clinical Toxicity Prediction Using Multi-Task Deep Neural Nets and Contrastive Molecular Explanations*. 2023.
8. Previously reviewed papers and reviews on Tox21, ToxCast, molecular fingerprints, GNNs, interpretable ML, ADMET and computational toxicology.

Prepared as an executable, stage-gated research blueprint. The numerical target is aspirational; all claims must follow the frozen leakage-safe protocol.